

Agentic AI for Earth Science Software: a field report from building Worldscope

David Matthew Hall · NVIDIA · Berlin, Germany

An interactive Earth-system workbench in which AI agents are not chatbots bolted on, but first-class users of every capability.



See it run
Source & walkthrough video

WHAT IT DOES

MODELS ON A GLOBE

Puts NVIDIA's **Earth2Studio** toolset on an interactive 3D globe alongside satellite imagery, climate archives, live weather feeds, and observational data.

Forecast, diagnostic, and assimilation runs; ensemble members, vertical levels, and time are all selectable axes.

FORMATS & GRIDS

Reads, writes, and visualizes **NetCDF**, **GRIB2**, **GeoTIFF**, and **Zarr**, across many grid types — including **HEALPix rendered natively**, rather than interpolated to a lat–lon raster first.

Every data display carries units and a legend.

AGENTS AS USERS

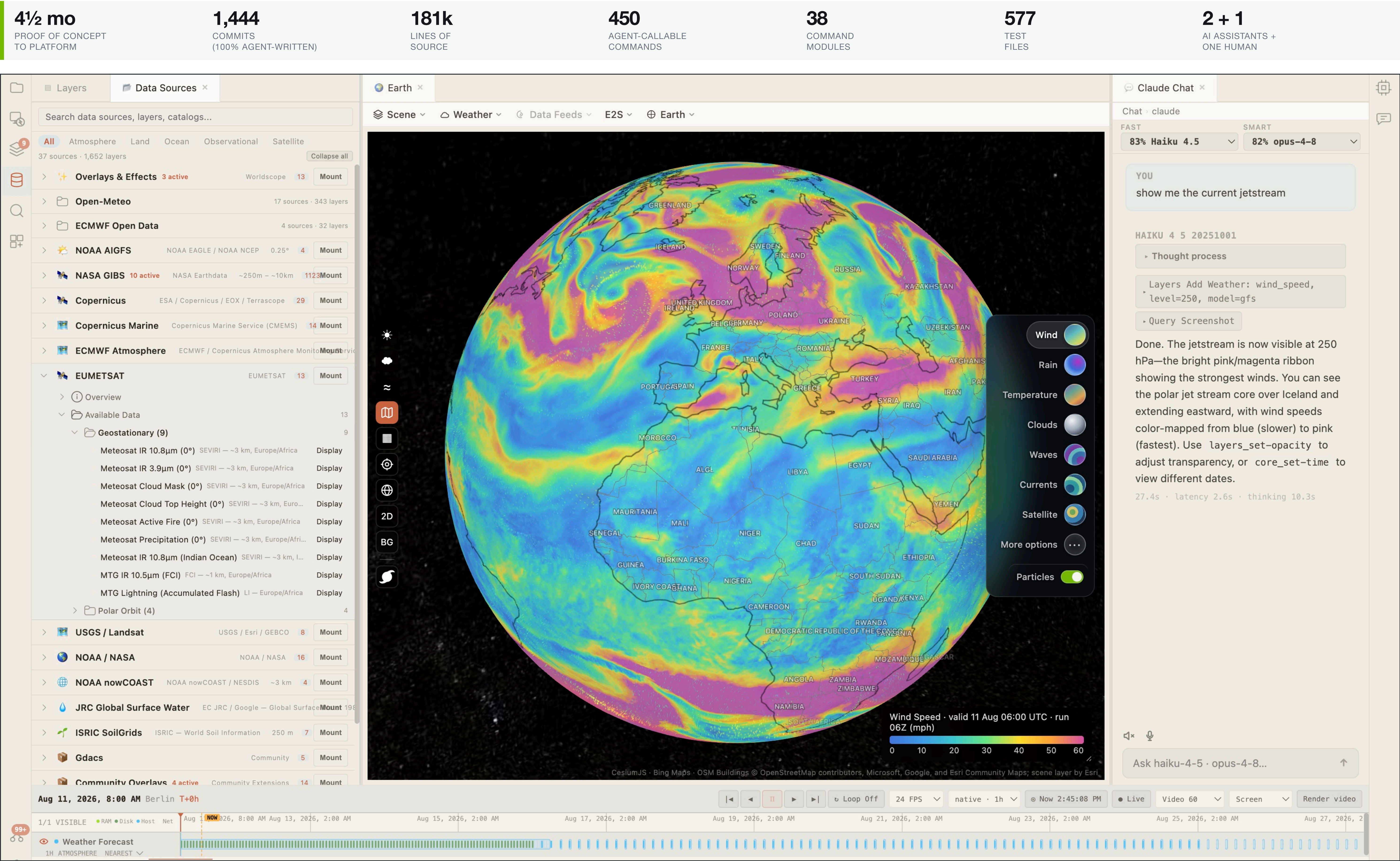
Every capability is exposed through the **Model Context Protocol** — not a chat window added afterwards.

An agent can load a forecast, select a variable, run a model, compare results, and manipulate the same visualization a human sees. **Multiple agents can work one scene at once.**

RUNS WHERE YOU ARE

Lightweight enough for any device, while taking advantage of whatever NVIDIA hardware is present.

With a single desktop **DGX Spark** it runs the vast majority of AI weather models **locally**. Intended for open release.



Worldscope. A global forecast field on the interactive globe, with the layer explorer on left and an agent session at right. The agent drives the same scene the human sees — same layers, same camera, same time cursor — through the Model Context Protocol, not a scripting sidecar.

One project is a data point. The pattern is the story.

1 The work did not vanish. It moved.

For well-specified features, agents produced implementations roughly **a hundred times faster** than I could have written them. That collapsed the cost of implementation and pushed the effort into specification, integration, and verification — none of which got cheaper.

4 Tests helped. Tests were not enough.

Agent-written tests caught regressions, but **could not establish that a complex interactive workflow behaved correctly end to end**. Plausible, runnable and quietly wrong is the failure mode that matters most in scientific software — and the one a green test suite does not catch.

OPEN QUESTIONS FOR THIS COMMUNITY

I WHERE THE WORK GOES

- What happens when AI agents are treated not as chatbots, but as *full participants* in an Earth-science workflow?
- If writing code becomes a hundred times cheaper, which part of the scientific workflow becomes the bottleneck — and is any group staffed for it?
- Is the durable artifact still the source code, or the *specification* that regenerates it?
- What is scientific programming a skill *in*, once the programming is free?

IV PROVENANCE & REVIEW

- If a published result rests on 180,000 lines no human has read line by line, what exactly was peer-reviewed?
- What should journals and programme committees require — prompts, model versions, tool versions, transcripts, seeds?
- Is agent-generated code reproducible once the agent that wrote it has been deprecated?
- Who is the author of a scientific tool, and what counts as a citable software contribution now?

II VERIFICATION & PHYSICALITY

- How do you verify software you did not write, at the rate it is now written?
- Who catches code that runs, passes its tests, and is *physically* wrong — flipped signs, unit slips, staggered-grid offsets, silent regridding, mishandled missing values, the wrong CRS?
- Generation scales; checking does not. Is that asymmetry the real ceiling on agentic science?
- What is the software equivalent of a physicality check — conservation, invariance, dimensional consistency, known-answer tests?

V THE GENERIC ATTRACTOR

- Agents regress toward the mean of their training data. Does that quietly pull the field toward common grids, common formats, and mainstream methods?
- How much of scientific novelty lives precisely in the choices an agent would talk you out of?
- Does agentic development accelerate science, or accelerate its *consolidation*?
- Do we get more science — or the same science, faster?

3 Verification was the hardest part.

Especially for interactive behavior, and partly for a structural reason: the agents observed the browser through **still screenshots** and had to infer what happened in between. Features routinely looked correct and did not work; fixing one silently broke another.

Where the effort went

Schematic — impressions from one project, not measurements.

WRITING IT MYSELF

Implementation	Spec	Integ.	Verify
----------------	------	--------	--------

DIRECTING AGENTS

Specification	Integration	Verification
---------------	-------------	--------------

■ Writing code ■ Saying what it must do ■ Making the parts cohere ■ Proving it is right

Answers welcome. Arguments preferred.

III EVALUATION & BENCHMARKS

- We benchmark the models. Should we benchmark the *code that produces them* — and what would that benchmark measure?
- Agents perceive interactive systems as still frames and infer the motion in between. What does that imply for evaluating anything time-dependent, interactive, or visual?
- Do agent-built pipelines quietly alter our datasets — resampling, gap-filling, reprojecting — in ways our metrics never see?
- If an agent writes the model, the training loop, *and* the evaluation, what independence is left?

VI PEOPLE, ACCESS, COMMONS

- If early-career scientists never write the code, where does the judgment to review it come from — and what do we train them to be good at?
- Does this widen the gap between well-resourced and under-resourced groups, or erase it?
- When a bespoke tool costs a weekend, do we still build shared community software — or does the toolchain quietly fragment?
- Is scientific software still worth *maintaining*, or is it now disposable and regenerable?

As the cost of intelligence drops rapidly toward zero, the scarce resource turns out to be **artistic vision and judgment**.

If that is right —
are we selecting for it, teaching it,
or rewarding it?